SSK Natural Gold — Architecture Audit & Migration Plan

System Architecture Audit, Codebase Inventory & Workflow Migration Specification

Role: Senior Full-Stack Software Architect & Technical Lead | **Date:** November 10, 2026 | **Status:** Read-Only Audit Completed

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1. Executive Summary

SSK Natural Gold is an enterprise e-commerce platform built for a startup retail edible oil enterprise in Tenkasi, Tamil Nadu. The company extracts and bottles 3 traditional cold-pressed (*Vaagai Marachekku*) oils: Traditional Groundnut Oil, Pure Sesame Oil with authentic Palm Jaggery (Karupatti), and Unrefined Coconut Oil.

The Simplified Workflow:
Homepage → Product Catalogue → Product Details → Cart → Customer Details → Submit Order Request → Admin Reviews Request → Admin Confirms Availability & Final Amount → Customer Pays via UPI QR → Admin Verifies Payment → Order Confirmation → Dispatch → Delivery.

Architectural Layer	Verified State	Audit Finding	Recommendation
Frontend SPA	React 18, Vite 6, Tailwind CSS	100% buildable; 27 views, 43 routes	Retain & simplify. Unhook auth & account views.
Global Stores	4 Zustand stores	useAppStore & useCartStore persist mock items	Clear pre-seeded cart items; support guest details.
Backend API	Node.js, Express, TS	Only GET /api/health implemented; 0 controllers	Implement request & notification endpoints.
Database / ORM	PostgreSQL, Prisma ORM 6.3.1	schema.prisma has 0 models; 0 migrations	Pristine slate. Define OrderRequest & Product models.

2. Verified Frontend Structure

The frontend is organized under frontend/src/ with modular feature folders: components/common (17 primitives), components/product (6 domain components), components/layout (10 shell layouts), stores (4 Zustand stores), mock (products, addresses, orders), and pages (27 view components).

Route Inventory & Target Classification:

Category / Route	Component	Target Classification	Action Description
/, /home	HomePage.tsx	Keep with modifications	Connect add-to-cart directly to useCartStore
/shop, /products	ShopAllPage.tsx	Keep unchanged	Full catalog filtering and price sorting
/products/:slug	ProductDetailPage.tsx	Keep with modifications	Direct 'Buy Now' to guest checkout form
/cart	CartPage.tsx	Simplify	Estimated subtotal; courier confirmed by admin
/checkout	CheckoutAddressPage.tsx	Keep with modifications	Convert to Guest Customer Details & Address Form
/checkout/payment	CheckoutPaymentPage.tsx	Remove / Transform	Transform into Order Request Review & Submit
/checkout/success	CheckoutSuccessPage.tsx	Keep with modifications	Order Request Received with Reference #
/track, /orders/:id	OrderTrackingPage.tsx	Keep with modifications	Guest lookup + dynamic UPI QR display
/login, /register, /account	Auth & Account views	Defer for future dev	Unhook from navigation; guest checkout model
/admin/orders	AdminOrdersPage.tsx	Keep with modifications	Central Admin Order Request & Fulfillment Desk

3. Verified Backend Structure

The backend under backend/src/ is currently a clean initial scaffold. Express is configured with CORS and JSON parsing. **Only 1 active endpoint exists:** GET /api/health. Prisma schema has zero models defined, and no database migrations have been executed. Controllers, services, and route definitions for orders and products are ready to be built directly for the simplified workflow.

4. Existing Functionality That Can Be Reused

- **Rich Product Specs:** Groundnut, Sesame (Karupatti), and Coconut oil datasets with Tamil names, FSSAI numbers, and pack sizes (500ml, 1L, 5L, 15L).
  - **Bilingual Engine:** Instant EN/TA language toggling across headers, cards, and footers.
  - **UI Design System:** 17 common components and 6 product components built with Tailwind CSS.
  - **Cart State Machine:** Zustand-based cart state management with reactive item counts and subtotals.
  - **Admin Layout Shell:** Responsive sidebar and header navigation for the operations console.

5. Screens That Need Modification & Deferrals

Modifications:

1. HomePage.tsx: Wire product cards directly to useCartStore.addItem().
2. CartPage.tsx: Remove pre-seeded items; display estimated subtotal; change CTA to 'Proceed to Customer Details'.
3. CheckoutAddressPage.tsx: Convert to Guest Customer Form (Name, WhatsApp, Address, Pincode).
4. CheckoutPaymentPage.tsx: Transform to Order Request Review & Submit view.
5. CheckoutSuccessPage.tsx: Display Unique Request Reference (e.g. SSK-REQ-XXXX) and timeline.
6. OrderTrackingPage.tsx: Public guest lookup by Reference # + Phone; display dynamic UPI QR when payment is pending.
7. AdminOrdersPage.tsx: Order review modal, shipping fee input, UPI verification, and AWB docket entry.

Deferred Features: Customer login/register suite, SMS OTP verification, customer account dashboard, automated payment gateway integration, and margin formula simulators.

6. Database Readiness & Proposed Schema

The proposed database schema for PostgreSQL via Prisma ORM establishes clean relational models with zero migration debt:

Model Name	Primary Fields	Purpose in Request Workflow
Product	id, slug, category, name, nameTa, fssaiLicense, primaryImage	Core oil variety metadata & storytelling
ProductVariant	id, productId, sku, volume, volumeMl, mrp, price, stock	Pack SKU inventory & server-side price truth
OrderRequest	id, orderReference, status, customerName, customerPhone, customerEmail, addressLine1, city, pincode, estimatedSubtotal, courierFee, finalPayableAmount, upiVpa, customerUtrNumber, courierPartner, awbNumber	Complete lifecycle record of the customer order request from submission to delivery
OrderItem	id, orderRequestId, variantId, productName, volumeLabel, unitPrice, quantity, totalPrice	Snapshot of ordered items and unit prices
OrderStatusLog	id, orderRequestId, status, note, createdByName, createdAt	Audit trail of status updates
AdminUser	id, email, name, passwordHash, isActive	Admin console authentication

7. Security & Business Logic Concerns

- **Server-Side Price Validation:** Subtotals must be calculated exclusively using database variant prices, completely ignoring client-submitted figures.
- **Database Persistence Prerequisite:** Every order request must be committed to PostgreSQL before WhatsApp/Email notifications are triggered.
- **Guest Order Privacy:** Public tracking queries require matching the phone number to return delivery addresses.
- **Duplicate UTR Checks:** Customer-submitted UPI transaction reference numbers must be checked for uniqueness across confirmed orders.
- **Admin Security:** Operations routes protected with secure session tokens; passwords hashed with bcrypt.

8. Recommended Implementation Sequence & Readiness

8-Phase Execution Blueprint:
Phase 1: Database Schema & Prisma Migration → **Phase 2:** Storefront Product & Order Request APIs → **Phase 3:** Admin Review & Payment APIs → **Phase 4:** Notification Services (WhatsApp & Email) → **Phase 5:** Frontend Guest Checkout Realignment → **Phase 6:** Frontend Admin Console Adaptation → **Phase 7:** Navigation & Route Streamlining → **Phase 8:** End-to-End Verification.

Overall Readiness Scorecard:

Pillar	Score	Technical Status
Frontend UI / UX	**9.5 / 10**	Production-ready Tailwind components, authentic branding, full bilingual support.
Domain Fit	**9.0 / 10**	Accurate cold-pressed oil specifications, Tamil names, and pack sizes.
State Management	**8.5 / 10**	Clean Zustand stores; ready to switch to guest session handling.
Backend Scaffold	**7.0 / 10**	Solid Express/TS base; needs domain routes and controllers.
Database Readiness	**6.0 / 10**	Clean slate with zero legacy migrations; ready for direct schema implementation.
Workflow Realignment	**9.5 / 10**	Request-based purchasing eliminates payment gateway friction and accelerates launch.
Overall Project Health	**8.5 / 10**	Ready for execution. Zero compile errors on frontend and backend.

Audit Status: Completed & Ready for Execution

This audit was conducted in strictly read-only mode. No source code was modified or deleted. The complete markdown specification is available in docs/SSK_Natural_Gold_Architecture_Audit_and_Migration_Plan.md. Standing by for user instruction to begin Phase 1: Database Schema & Prisma Migration.